

Network **Protocols** Stack

Application Layer (L5-L7)



Web

HTTP(S)

WebSocket

REST/GrahpQL



Security

TLS/SSL

HTTPS

OAuth/SAML



Multimedia

RTMP/RTSP

SIP/SDP

HLS/DASH



Name

DNS

mDNS

DNSSEC



Mail

SMTP

POP3/IMAP

DMARC/SPF



Transfer

FTP/FTPS

SFTP

SMB/CIFS



Remote

SSH

RDP

VNC

Transport Layer (L4)



Reliable Transport

TCP

SCTP

QUIC

RDMA

mTLS

HTTP/2

HTTP/3

TLS 1.3

SSH

WEbTrasport



Unreliable Transport

UDP

RTP

RTCP

DCCP

SRTP

DTLS

CoAP

DNS

TFTP

mDNS

Network Layer (L3)



Core IP

IPv4

IPv6



Routing

RIP

OSPF

BGP

IS-IS



Advanced
Routing

MPLS

SRv6

VRRP

LISP



Security

IPsec

IKE

GRE

VPN



Mail

ICMP

ICMPv6

SNMP

NTP

DHCP

DHCPv6

NetFlow

sFlow

Data Link Layer (L2)



Ethernet

802.3

Jumbo Frames



VLAN

802.1Q

QinQ



WiFi

802.11

a/b/g/n/ac/ax



Security

802.1X

MAC Sec



L2 Addr

ARP

NDP



Switching

STP/RSTP

LACP



WAN Technologies

PPP

HDLC

MPLS

ATM

Frame

xDSL